



ELECTRICAL MEASUREMENTS

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ELECTRICAL MEASUREMENTS

Scope of presentation

Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads



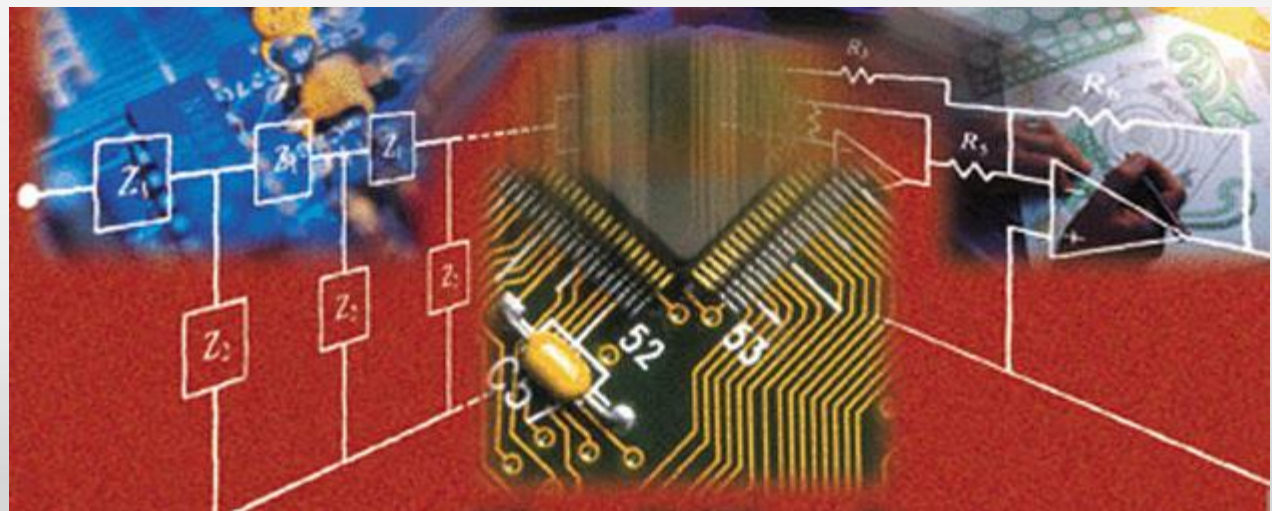
Menoufia University FACULTY OF Electronic Eng

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors



Thanks & Questions

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Thanks & Questions

1. System disturbance due to measurement.
✓ Measurements in electric circuits

Example 3.1: Suppose that the components of the circuit shown in Figure 3.1(a) have the following values:

$$R_1 = 400 \, \Omega; R_2 = 600 \, \Omega; R_3 = 1000 \, \Omega; R_4 = 500 \, \Omega; R_5 = 1000 \, \Omega$$

The voltage across AB is measured by a voltmeter whose internal resistance is 9500.

What is the measurement error caused by the resistance of the measuring instrument?

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1. System disturbance due to measurement.
✓ Measurements in electric circuits

Solution

Proceeding by applying Th'evenin's theorem to find an equivalent circuit to that of Figure 3.1(a) of the form shown in Figure 3.1(b), and substituting the given component values into the equation for R_{AB} (3.1), we obtain:

$$R_{AB} = \frac{[(1000^2/2000) + 500]1000}{(1000^2/2000) + 500 + 1000} = \frac{1000^2}{2000} = 500 \Omega$$

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1. System disturbance due to measurement.
✓ Measurements in electric circuits

Solution

$$\frac{E_m}{E_0} = \frac{R_m}{R_{AB} + R_m}$$

$$E_0 - E_m = E_0 \left(1 - \frac{R_m}{R_{AB} + R_m} \right)$$

$$E_0 - E_m = E_0 \left(1 - \frac{9500}{10\,000} \right) = 0.95E_0$$

Thus, the error in the measured value is 5%.

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2. Errors due to environmental inputs

- ✓ An environmental input is defined as an apparently real input to a measurement system that is actually caused by a change in the environmental conditions surrounding the measurement system.
- ✓ The fact that the static and dynamic characteristics specified for measuring instruments are only valid for particular environmental conditions (e.g. of temperature and pressure).
- ✓ These specified conditions must be reproduced as closely as possible during calibration exercises because, away from the specified calibration conditions, the characteristics of measuring instruments vary to some extent and cause measurement errors.

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2. Errors due to environmental inputs

- ✓ In any general measurement situation, it is very difficult to avoid environmental inputs, because it is either impractical or impossible to control the environmental conditions surrounding the measurement system.
- ✓ System designers are therefore charged with the task of either reducing the susceptibility of measuring instruments to environmental inputs or, alternatively, quantifying the effect of environmental inputs and correcting for them in the instrument output reading.
- ✓ The techniques used to deal with environmental inputs and minimize their effect on the final output measurement follow a number of routes as discussed below.

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3. Wear in instrument components.

- ✓ Systematic errors can frequently develop over a period of time because of wear in instrument components.
- ✓ Recalibration often provides a full solution to this problem.

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4. Connecting leads.

- ✓ In connecting together the components of a measurement system, a common source of error is the failure to take proper account of the resistance of connecting leads (or pipes in the case of pneumatically or hydraulically actuated measurement systems).
- ✓ Therefore, careful consideration needs to be given to the choice of connecting leads.
- ✓ Not only should they be of adequate cross-section so that their resistance is minimized,
- ✓ but they should be adequately screened if they are thought likely to be subject to electrical or magnetic fields that could otherwise cause induced noise.

Questions and Discussion

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End of Lecture

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Good Luck
See You in
Next Lecture . . .



With My Best Wishes
Dr. Eng. Mohamed Abdo